

In Angular, **SCSS variables are compile-time only** — you cannot directly change them at runtime using `ViewChild` because SCSS is preprocessed into plain CSS before the app runs.

If you want to dynamically change a style value for an element at runtime, you should use **CSS custom properties (variables)** instead, because they are supported natively by browsers and can be updated via JavaScript or Angular.

✓ Example: Change a CSS variable of a class using `ViewChild`

1. SCSS (use CSS variables inside your class)

```
/* styles.scss or component.scss */
.my-box {
  --box-color: red; /* default value */
  background-color: var(--box-color);
  width: 200px;
  height: 200px;
}
```

2. Angular Component

```
import { Component, ElementRef, ViewChild, AfterViewInit } from '@angular/core';

@Component({
  selector: 'app-root',
  template: `
```

```

    <div #box class="my-box"></div>
    <button (click)="changeColor('blue')">Make Blue</button>
    <button (click)="changeColor('green')">Make Green</button>
  `,
  styleUrls: ['./app.component.scss']
})
export class AppComponent implements AfterViewInit {
  @ViewChild('box') boxRef!: ElementRef<HTMLDivElement>;

  ngAfterViewInit() {
    // Optional: set initial value dynamically
    this.changeColor('orange');
  }

  changeColor(color: string) {
    if (this.boxRef?.nativeElement) {
      this.boxRef.nativeElement.style.setProperty('--box-color', color);
    }
  }
}

```

3. How it works

- **SCSS variable** (\$var) → compile-time only, cannot be changed at runtime.
- **CSS variable** (--var) → runtime changeable via `element.style.setProperty`.

- `ViewChild` gives you a reference to the DOM element so you can update its CSS variable.

4. Benefits

- No need to recompile SCSS.
- Works per element (scoped changes).
- Can be animated with CSS transitions.

 **Tip:** If you want to change the variable for the whole app, you can set it on `document.documentElement` instead of a single element:

```
document.documentElement.style.setProperty('--box-color', 'purple');
```

If you want, I can also show you **a version that changes the CSS variable for all elements with a given class**, not just the one from `ViewChild`.

Do you want me to prepare that?